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System and method to determine engine thrust of a taxiing aircraft

Abstract

A computing device to determine engine thrust of an aircraft during taxiing. The computing device includes memory circuitry configured to store computer-readable program code. Processing circuitry is configured to execute the computer-readable program code to cause the computing device to: identify an aircraft on an airport surface; calculate an acceleration of the aircraft during the taxiing of the aircraft; compare the acceleration with predetermined data from similar aircraft; and based on the comparison, determine an engine thrust for the aircraft while the aircraft is taxiing.

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Background/Summary

RELATED APPLICATIONS

(1) This claims priority to European Patent Application No. EP 22386007.3, filed Feb. 16, 2022, the disclosure of which is incorporated herein by reference in its entirety.

TECHNOLOGICAL FIELD

(2) The present disclosure relates generally to monitoring aircraft on an airport surface and, more particularly, to determining an engine thrust of an aircraft that is moving taxiing on an airport surface.

BACKGROUND

(3) Aircraft taxi on airport surfaces using engine thrust created by the engines. The engines of the aircraft can be set to different thrust levels to provide different engine thrust to the aircraft. In one example, the engines can be set to one of a first thrust level or a second reduced thrust level. The first thrust level provides for a higher acceleration and velocity during taxiing. The second thrust level is a reduced thrust level that results in slower acceleration and velocity. The reduced thrust level of each engine can vary with one setting including a zero thrust level in which the engine produces no force to move the aircraft. For aircraft operating with one or more of the engines at a zero thrust level, the overall engine thrust of the aircraft is provided by one or more of the other engines operating at a non-zero thrust level.

(4) Pilots taxi with the engines at different thrust levels for a variety of reasons. A higher engine thrust provides for higher acceleration and velocity which can lessen commute times between the gate and the runway or vice versa. Lower thrust levels may conserve fuel and may also result in lower emissions.

(5) There are advantages in knowing the thrust level of a taxiing aircraft. As stated above, these levels can be used to more accurately determine fuel usage and/or emissions. Knowing the thrust level of an aircraft can also facilitate scheduling at an airport. An air traffic controller responsible for coordinating multiple aircraft at an airport would be better able to efficiently schedule aircraft at the airport if they knew the thrust levels and thus the acceleration and velocity of the aircraft. A gate operator responsible for preparing the gate for an incoming flight would be better able to make certain the gate was prepared on time if they more precisely knew the expected arrival at the gate.

(6) Pilots are not required to report the thrust levels used during taxiing and there is not a system or manner of determining the engine thrust of an aircraft. Thus, there is a need to determine the engine thrust of an aircraft that is taxiing on an airport surface.

SUMMARY

(7) One example is directed to a computing device to determine engine thrust of an aircraft during taxiing. The computing device comprises memory circuitry configured to store computer-readable program code. Processing circuitry is configured to execute the computer-readable program code to cause the computing device to: identify an aircraft on an airport surface; calculate an acceleration of the aircraft during the taxiing of the aircraft; compare the acceleration with predetermined data from similar aircraft; and based on the comparison, determine an engine thrust for the aircraft while the aircraft is taxiing.

(8) Comparing the acceleration with the predetermined data from similar aircraft may comprise comparing the acceleration and the time period with acceleration data and time period data for other aircraft each with a common manufacturer and model.

(9) The processing circuitry may be further configured to determine the number of engines on the aircraft based on identified model of the aircraft.

(10) For each of the engines the processing circuitry may be configured to determine whether each of the engines is operating at a higher thrust level or a lower thrust level.

(11) The lower thrust level may be a zero thrust level.

(12) The processing circuitry may be configured to: determine a starting point of the aircraft on the airport surface when the aircraft is stationary; determine an ending point that is positioned away

from the starting point with the aircraft moving along the airport surface at the ending point; and calculate the acceleration based on a change in velocity between the starting and ending points and the time period for the aircraft to move between the starting and ending points.

(13) The processing circuitry may be configured to determine that the acceleration falls within a cluster of data points in the predetermined data and determine the engine thrust based on the cluster.

(14) The processing circuitry may be configured to determine a number of the engines and a thrust level of each of the engines.

(15) The predetermined data may be stored at the monitoring node.

(16) One example is directed to a computing device to determine engine thrust of an aircraft during taxiing. The computing device comprises memory circuitry configured to store computer-readable program code. Processing circuitry is configured to execute the computer-readable program code to cause the computing device to: identify the aircraft by model; identify a direction of travel of the aircraft; determine an acceleration of the aircraft over a time period during taxiing of the aircraft; compare the acceleration and the time period with predetermined data of other aircraft of the same model and the same direction of travel; determine that the acceleration and time period is within a cluster of data points in the predetermined data with the cluster corresponding to an engine thrust for the aircraft; and determine that the aircraft is taxiing with a same number of engines that are operating at same thrust levels as other aircraft from the corresponding cluster.

(17) The processing circuitry may determine an engine thrust of the aircraft.

(18) The processing circuitry may calculate the acceleration between a first point where the aircraft is stationary and a second point where the aircraft is moving.

(19) The processing circuitry may calculate the acceleration during the time period with a velocity of the aircraft constantly increasing.

(20) The computing device may be further configured to calculate the acceleration and the time period of the acceleration of the other aircraft and store the thrust levels as the data points in the predetermined data.

(21) One example is directed to a method of calculating engine thrust of an aircraft that is taxiing along an airport surface of an airport. The method comprises: identifying an aircraft that is on an airport surface; calculating an acceleration over a period of time of the aircraft taxiing along the airport surface; comparing the acceleration and the time period to data points from other ones of the same aircraft that each had an acceleration over a time period during taxiing; determining that the acceleration and the time period corresponds to a cluster of the data points with the cluster corresponding to a number of engines operating at one or more thrust levels; and determining that the aircraft is taxiing with the same number of engines and the same thrust levels as the cluster.

(22) Identifying the aircraft may comprise identifying a model of the aircraft from records maintained at the airport.

(23) Calculating the acceleration of the aircraft may comprise determining a change in velocity of the aircraft between first and second points along the airport surface relative to the time period for the aircraft to move between the first and second points with the velocity of the aircraft at the first point being zero.

(24) The method may further comprise obtaining the data points from one or more remote entities that monitor other aircraft.

(25) The method may further comprise that each of the one or more engines on the aircraft has a full thrust level or a zero thrust level.

(26) The method may further comprise obtaining the data points from previous aircraft while taxiing at the airport.

(27) The features, functions and advantages that have been discussed can be achieved independently in various examples or may be combined in yet other examples, further details of which can be seen with reference to the following description and the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of an aircraft.
- (2) FIG. 2 is a schematic diagram of an airport.
- (3) FIG. 3 is a schematic diagram of an aircraft that moves along an airport surface between first and second positions.
- (4) FIG. 4A is a time and acceleration chart of data points of predetermined data for aircraft that include two engines.
- (5) FIG. 4B is a time and acceleration chart of data points of predetermined data for aircraft that include three engines.
- (6) FIG. 5 is a flowchart diagram of a method of determining engine thrust of an aircraft during taxiing on an airport surface.
- (7) FIG. 6 is a flowchart diagram of a method of determining engine thrust of an aircraft during taxiing on an airport surface.
- (8) FIG. 7 is a flowchart diagram of a method of determining engine thrust of an aircraft during taxiing on an airport surface.
- (9) FIG. 8 is a schematic diagram of a monitoring node.
- (10) FIG. 9 is a diagram of entities that can communicate with a monitoring node.
- (11) FIG. 10 is a flowchart diagram of a method of building a database of predetermined data corresponding to aircraft taxiing on an airport surface.

DETAILED DESCRIPTION

- (12) FIG. 1 illustrates an example of an aircraft **20** that includes engines **21**. The aircraft **20** can include different numbers of engines **21**, with FIG. 1 including an example with two engines **21**. The engines **21** are configured to provide thrust to propel the aircraft **20** during flight, as well as propel the aircraft during taxiing on an airport surface prior to and after the flight. Various types of engines **21** can be used on the aircraft **20**, including but not limited to turboprop engines, turbojet engines, turboshaft engines, turbofan engines, and ramjet engines. The aircraft **20** can be configured to transport various cargo, such as passengers and/or and freight. The aircraft **20** further includes landing gear **22** (see FIG. 3) that includes tires for movement on the airport surface. The landing gear **22** is deployed while the aircraft **20** is on the ground and can be retracted during flight.
- (13) FIG. 2 illustrates an airport **100** from which the aircraft **20** can takeoff and land. The airport **100** includes airport surfaces **101** along which the aircraft **20** moves while on the ground. The airport surfaces **101** can include but are not limited to runways **102** and taxiways **103**. The taxiways **103** extend between the runways **102** and the gates **104** at the terminal **105**.
- (14) The aircraft **20** taxis along the airport surfaces **101** when on the ground. In some examples, the taxiing includes moving along the taxiways **103** between the gates **104** and the runways **102**. The taxiing can also occur along the runways **102**, such as moving along a section of the runway **102** after a landing is completed or when crossing one runway intersection in order to reach another runway **102**. In some examples, the aircraft **20** lands on a runway **102** and completes the landing while still along a central section of the runway **102**. The taxiing includes moving along the remainder of the runway **102** and then along a taxiway **103**.
- (15) The aircraft **20** is propelled along the ground during taxiing by the one or more of the engines **21**. The acceleration and velocity of the aircraft **20** during taxiing depends upon the overall engine thrust provided by the one or more engines **21**. An aircraft **20** with a higher overall engine thrust moves at a higher acceleration and velocity than a corresponding aircraft **20** with a lower overall engine thrust.
- (16) In some examples, each of the engines **21** operates at one of two thrust levels. The thrust levels include a first higher level and a second lower level. In some examples, the thrust levels include a

full thrust level and a zero-thrust level. The lower thrust level can be used to conserve fuel and can further provide environmental benefits by lessening the amount of exhaust produced by the engine **21**. In other examples, the lower thrust level is above zero and provides for propelling the aircraft **20** at a slower rate. In other examples, the engines **21** operate at one or more intermediate thrust levels in addition to the high and low thrust levels. In other examples, the high thrust level is less than a full thrust level for the engine **21**.

(17) A monitoring node **110** monitors the aircraft **20** during taxiing to determine the thrust levels of the one or more engines **21** and thus the overall engine thrust of the aircraft **20**. The monitoring node **110** can be associated with various entities **80**. Some examples includes the airport **100**, such as through a control tower that monitors the overall operation of the airport **100**. Other examples include an airline that schedules the arrival and departure of aircraft **20**. Other examples include a flight tracker (e.g., Flight Radar **24**, Flight Aware, Spire) that monitors the status of flights. Other examples include a governmental unit (e.g., Federal Aviation Administration (FAA)) that regulates various aspects of aviation. In other examples, the monitoring node **110** is independent and is not associated with any other entity **80**.

(18) The monitoring node **110** calculates the thrust levels of the engines **21** of an aircraft **20** using the acceleration of the aircraft **20** moving along the airport surface **101** and the amount of time of the acceleration. This is schematically illustrated in FIG. **3** with the monitoring node **110** monitoring the movement of the aircraft **20** along the airport surface **101** from a first point A to a second point B. The acceleration of the aircraft **20** is calculated for the section of the airport surface **101** between the points A, B based on the following formula:

$$\text{acceleration} = \Delta \text{velocity} / \Delta \text{time} \quad \text{Eq. 1}$$

(19) For example, the aircraft **20** is monitored moving at a velocity of 10 m/s at point A and a velocity of 25 m/s at point B. The aircraft **20** covers the distance between points A and B in 2 seconds. The acceleration is 7.5 m/s.². In one example, the acceleration calculation occurs from a point on the airport surface **101** where the aircraft is stationary (i.e., point A has a velocity=0). For example, from a point on the taxiway **103** where the aircraft **20** paused after completing a landing, with the taxiing calculation occurring once the aircraft **20** begins to move again along the runway **102**.

(20) In some examples, the acceleration of the aircraft **20** is calculated over a distance in which the velocity of the aircraft **20** is changing. For example, the velocity of the aircraft **20** is increasing between points A and B. Distances in which the aircraft **20** has a constant or substantially constant velocity are not used for the acceleration calculation. For example, an aircraft **20** is monitored moving along an airport surface **101**. For a first section between points A and B, the aircraft **20** accelerates. For a second section between points B and C, the aircraft **20** has a constant or substantially constant velocity with no additional acceleration. The acceleration of the aircraft **20** is determined using the data between points A and B, with the movement between points B and C not included in the calculations.

(21) In addition to calculating the acceleration, the monitoring node **110** identifies the aircraft **20**. This can include identifying one or more of the manufacturer of the aircraft **20**, the model, and one or more variants (e.g., Boeing 747-8, Boeing 737 MAX, Boeing 787, Boeing 777-X). Identification of aircraft provides for determining the number of engines **21** on the aircraft **20**. The identification also provides for comparing the aircraft **20** with data from other similar aircraft **20** to determine the engine thrust as will be explained below.

(22) The monitoring node **110** also determines whether the aircraft **20** is taxiing outward in preparation for takeoff or taxiing inward after landing. The direction of taxiing can affect the acceleration because of the amount of fuel on the aircraft and resultant mass. In some examples, the aircraft **20** has a lower acceleration during taxiing outward because of the larger fuel load than a comparable aircraft **20** that is taxiing inward that has less fuel due to fuel usage during the flight.

(23) The monitoring node **110** determines the engine thrust of the aircraft **20** by comparing the

acceleration data with other similar taxiing aircraft **20**. In some examples, the comparison uses other aircraft **20** of the same manufacturer and model and that were taxiing the same direction. Other examples include other similar aircraft **20** of the same make and model regardless of direction. Other example include other aircraft **20** with the same number of engines **21**.

(24) FIG. 4A illustrates predetermined data used by the monitoring node **110** to compare with the current aircraft **20**. The predetermined data are from other aircraft **20** that have been observed taxiing on the airport surface. The predetermined data are for aircraft **20** with one or more of the same characteristics as the monitored aircraft **20**. For example, the predetermined data are from numerous other aircraft **20** of the same make and model that were moving in the same direction.

(25) As illustrated in FIG. 4A, the predetermined data includes a number of data points **41** that each represent taxiing data for a previously recorded aircraft **20**. Each data point **41** can be plotted on a time-acceleration graph. The data points **41** group into one or more clusters **40** that correspond to the number of engines **21** on the aircraft **20**. For example, FIG. 4A includes two clusters **40a**, **40b** as the data are for an aircraft **20** having two engines **21**. Other examples include three clusters **40** for aircraft **20** with three engines **21**, four clusters **40** for four engines **21**, etc.

(26) The monitoring node **110** compares the acceleration data from the current aircraft **20** with the predetermined data and determines whether the monitored aircraft **20** falls into one of the clusters **40**. Using the example of FIG. 4A as an example, the monitoring node **110** determines whether data point of the monitored aircraft **20** falls into either of clusters **40a** or **40b**. A data point **41** that falls into the first cluster **40a** includes a lower acceleration and a higher time. This cluster **40a** corresponds to aircraft **20** that were taxiing with one engine **21** operating at a first thrust level (e.g., a high thrust level) and a second engine operating at a second thrust level (e.g., a low thrust level). If the monitoring node **110** determines the data point **41** falls into this cluster **40a**, the monitoring node **110** determines that the monitored aircraft **20** is operating in a similar manner with the engines **21** at the same thrust levels. If the monitoring node **110** determines that the data point **41** falls into the second cluster **40b** that corresponds to aircraft **20** that operated with both of the engines **21** at a high thrust level, the monitoring node **110** determines that the monitored aircraft **20** is also operating at this same engine thrust.

(27) The monitoring node **110** can use one or more different algorithms to place the data points **41** of the predetermined data into the one or more clusters **40**. The monitoring node **110** can also use the manufacturer and model of the aircraft **20** to determine the number of engines **21** and thus the number of clusters **40** into which to segregate the data. One algorithm includes connectivity models in which the data points **41** are classified into separate clusters **40** and then aggregated as the distance between the data points **41** decreases. Connectivity models can also classify the data points **41** in a single group and then partition the group into separate clusters **40** as the distance increases. Another connectivity model algorithm utilizes iterative clustering algorithms in which the similarity between the data points **41** is based on the closeness of data points **41** to a centroid of the clusters **40**. Another algorithm uses distribution models based on the probability that all data points **41** in the cluster **40** belong to the same distribution. Another algorithm uses density models that identify the data space for areas of varied density of data points **41** and isolates and assigns the different density regions into the same cluster **40**.

(28) Once the data points **41** of the predetermined data are grouped into the one or more clusters **40**, the monitoring node **110** determines whether the data point **41** of the monitored aircraft **20** falls into one of the clusters **40**. One clustering algorithm includes soft clustering in which the data point **41** is assigned a probability of being within the clusters **40**. Another algorithm uses hard clustering in which the data point **41** is assigned to a cluster **40**.

(29) As previously stated, the number of clusters **40** in the predetermined data corresponds to the number of engines **21** on the aircraft **20**. FIG. 4B illustrates an example of predetermined data for a three-engine aircraft **20**. Data points **41** form clusters **40a**, **40b**, **40c**. Cluster **40a** indicates one engine **21** at a high thrust level and two engines **21** at a low thrust level (e.g., one engine **21** at a full

thrust level or some positive thrust level and two engines at a low thrust level). Cluster **40b** indicates two engines **21** at the high thrust level and one engine at the low thrust level. Cluster **40c** indicates all three engines **21** at the high thrust level.

(30) The monitoring node **110** determines the engine thrust of the aircraft **20** by determining the thrust level of the engines **21** based on a comparison with previous aircraft **20**. FIG. 5 illustrates one method of determining the engine thrust that includes that the monitoring node **110** identifies the aircraft **20** (block **150**) and calculates the acceleration of the aircraft **20** over a time period as the aircraft **20** moves along the airport surface **101** (block **152**). The monitoring node **110** compares these data of the monitored aircraft **20** with predetermined data for similar aircraft **20** (block **154**). Based on the comparison, the monitoring node **110** determines a thrust level for each of the engines **21** on the aircraft **20** and thus an engine thrust for the aircraft **20** (block **156**).

(31) The engines **21** of the aircraft **20** are powered to propel the aircraft **20** during taxiing along the airport surface **101**. In some examples, the thrust level for each of the engines **21** includes either providing no thrust or providing a full amount of thrust. In other examples, one or more of the engines **21** can provide differing amounts of thrust between the no thrust and full thrust levels. Using the example of FIG. 4A, the data points **41** within cluster **40b** include previous aircraft **20** taxiing using some amount of thrust on both engines **21**. Data points **41** within the second cluster **40a** include previous aircraft **20** that taxied using one engine **21** with some amount of thrust and a second engine **21** with no thrust. Data points **41** that do not fall within either cluster **40** are outlier data. These outlier data points **41** can be caused by aircraft **20** using differing amounts of thrust on one or both engines **21** (e.g., $\frac{3}{4}$ engine thrust, $\frac{1}{2}$ engine thrust). The outlier data points **41** can also be a result of different fuel amounts in the aircraft **20** than normal for other similar aircraft **20** thus causing faster or slower taxiing. These outlier data points can also be caused by unusual weather conditions, such as fog, rain, and snow that require pilots to operate the aircraft **20** at slower speeds.

(32) FIG. 6 illustrates functionality of the monitoring node **110** in determining the engine thrust of an aircraft **20**. The monitoring node **110** identifies the aircraft (block **160**) and calculates acceleration for the aircraft **20** over a time period while taxiing along the airport surface **101** (block **162**). The monitoring node **110** compares the acceleration and time period with predetermined data for other similar taxiing aircraft **20**. The comparison determines whether the acceleration data falls within one of the clusters **40** in the predetermined data (block **164**). When the monitoring node **110** determines that the acceleration data fall within one of the clusters **40**, the monitoring node **110** determines that the aircraft **20** is taxiing with corresponding thrust levels of the engines **21** (block **166**). When the acceleration data do not correspond to one of the clusters **40**, the monitoring node **110** determines that there is a different hybrid engine use (block **168**). The different engine use can be caused by one or more factors including but not limited to differing amounts of fuel in the aircraft **20**, different thrust levels from one or more of the engines **21**, and weather conditions.

(33) FIG. 7 illustrates functionality of the monitoring node **110** to determine an engine thrust of an aircraft **20**. The monitoring node **110** identifies the aircraft **20** by model (block **180**). The model provides the monitoring node **110** with the number of engines **21** on the aircraft **20** and can also include various other aspects, including but not limited mass of the aircraft **20**, total engine thrust, engine thrust levels for the engines **21**, and acceleration ranges for the aircraft **20**. The monitoring node **110** determines a direction of travel of the aircraft **20** (block **182**). The direction of travel includes taxiing outward away from the gate **104** in preparation for takeoff and taxiing inward towards the gate **104** after landing. Outward taxiing includes the aircraft having a larger weight and mass as the aircraft **20** is fueled in preparation for the flight. Inward taxiing includes a smaller fuel load as a portion of the fuel has been consumed during the flight. Outward taxiing can be slower due to the weight of the extra fuel.

(34) The monitoring node **110** determines an acceleration of the aircraft **20** (block **184**). The acceleration includes the change in velocity of the aircraft **20** over a time period. In some examples,

the acceleration is determined over a time period from when the aircraft **20** is initially stationary to a time in which the aircraft **20** has a given velocity. In some examples, the acceleration is determined only during the time period in which the aircraft **20** is accelerating.

(35) The acceleration and time period are then compared to predetermined data for other aircraft **20** with the same model number and manufacturer and moving in the same direction (block **186**). The predetermined data include one or more clusters **40** of data points **41** that each correspond to a different number of engines **21** providing the engine thrust. The data point **41** calculated for the acceleration and direction of the aircraft **20** is determined to fall within one of the clusters **40** (block **188**). The thrust levels of the one or more engines **21** of the aircraft **20** is determined to be the same as that of the cluster **40** (block **190**). The engine thrust information can include a total engine thrust of the aircraft **20**. The engine thrust information can also include the thrust level for each of the engines **21** on the aircraft **20**.

(36) In the various examples, the order in which the methodology proceeds can vary. In some examples, the monitoring node **110** initially identifies the aircraft **20** prior to calculating the acceleration data. In other examples, the identification and calculation occur concurrently.

(37) As illustrated in FIG. **8**, the monitoring node **110** includes a computing device **119** with processing circuitry **111** that include one or more microprocessors, microcontrollers, Application Specific Integrated Circuits (ASICs), or the like, configured with appropriate software and/or firmware. A computer readable storage medium (shown as memory circuitry **112**) stores data and computer readable program code **115** that configures the processing circuitry **111** to implement the techniques using algorithms and logic described above. Memory circuitry **112** is a non-transitory computer readable medium and can include various memory devices such as random access memory, read-only memory, and flash memory. Communications circuitry **113** connects the monitoring node **110** to one or more other entities **80** as will be explained below. The communications can be through one or more of a cellular network, satellite communication network, and a packet data network. The communications circuitry **113** can support a wired connection (e.g., Ethernet), a wireless connection, or both. A database **114** is stored in a non-transitory computer readable storage medium (e.g., an electronic, magnetic, optical, electromagnetic, or semiconductor system-based storage device). The database **114** can be local or remote relative to the monitoring node **110**.

(38) In some examples, the predetermined data are maintained in the memory circuitry **112** and/or the database **114**. In other examples, the memory circuitry **112** is maintained remotely from the monitoring node **110** and accessed to calculate the engine thrust of a monitored aircraft **20**.

(39) The monitoring node **110** communicates with one or more entities **80**. The communication can provide for one or more of the aspects about the movement of the aircraft **20** on the airport surface **101** and the predetermined data about similar aircraft **20**. FIG. **9** illustrates examples of the different entities **80** that can communicate with the monitoring node **110**. Entities **80** include but are not limited to: the aircraft **20** itself, the airport **100** (e.g., the control tower that is monitoring aircraft **20**), an airline **81** (e.g., Delta Airlines, United Airlines) of which the aircraft **20** is part of a larger fleet, a flight tracker **82** (e.g., Flight Aware **24**, Flight Radar, Inspire), a government entity **83** (e.g., Federal Aviation Administration (FAA)), and an aircraft manufacturer **84** (e.g., Boeing). One or more of these entities **80** can provide information about one or more aspects of the aircraft **20** such as but not limited to acceleration, velocity, time period of movement, direction of travel, and manufacturer and model number. The monitoring node **110** itself may also be positioned at the airport **100** to collect the needed data to calculate the engine thrust.

(40) In some examples, the monitoring node **110** identifies the aircraft **20**, the direction of travel, and movement data to calculate the acceleration including velocity and time period from one or more of the entities **80**. In some examples, the data regarding movement of the aircraft **20** on the airport surface **101** is obtained by one or more flight trackers **82**. The flight trackers **82** provide the locations of the aircraft **20** along the airport surface **101**, velocity, direction of travel, and time

periods of the aircraft **20** while taxiing along the airport surface **101**. The flight trackers **82** can also provide the identity of the aircraft **20**.

(41) In some examples, the monitoring node **110** includes one or more personnel that monitor the incoming and outgoing flights and enters the information into the computing device **119**. The data can include one or more of the identity of the aircraft **20** and the movement data to calculate the acceleration data.

(42) The data points **41** of the predetermined data can be established in different manners. One example is illustrated in FIG. **10** that includes the monitoring node **110** building the predetermined data. Initially the monitoring node **110** has no or little previous data of the movement of previous aircraft. To build these data, the monitoring node **110** identifies aircraft **20** moving on the airport surface (block **300**) and a direction that the aircraft **20** are traveling (block **302**). The monitoring node **110** calculates the acceleration of the aircraft **20** over a time period (block **304**) and stores the data (block **306**). The monitoring node **110** repeats the process to build the database of predetermined data. In some examples, the monitoring node **110** builds the database until obtaining a predetermined amount of data points have been calculated. Once this amount of data is obtained, the monitoring node **110** is able to calculate engine thrust for subsequently monitored aircraft **20**.

(43) In some examples, the monitoring node **110** builds the database exclusively from aircraft **20** taxiing at a particular airport **100**. The monitoring node **110** maintains different groupings of data for aircraft **20** at different airports **100**. In other examples, the monitoring node **110** combines data for aircraft from two or more different airports **100**. The assumption with this collection method is that an aircraft **20** of the same model and manufacturer has the same acceleration at different airports **100**.

(44) In other examples, data points **41** can be obtained from the manufacturer **84** of the aircraft **20**. These data can be from testing performed on the aircraft **20** and/or different testing algorithms and/or simulations run on the technical specifications of the aircraft **20**. In other examples, the data are obtained from a flight tracker **82** that monitors aircraft **20** at numerous airports **100** around the world. In yet other examples, the data are built using a combination of two or more of these sources of data.

(45) The acceleration data calculated by the monitoring node **110** can be supplied to one or more of the entities **80**. In some examples, the interested entities **80** subscribe to the monitoring node **110** to have access to the data. The entities **80** can use their devices to access the monitoring node **110** through a network (e.g., Internet). In some examples, one or more of the entities **80** can use his or her respective device to access the monitoring node **110** through a separate portal. Each entity's portal can include a secure interface through which the entity **80** can access the information that is assigned to them. In some examples, the data are divided with the different data available to a limited number of the entities **80**.

(46) The entities **80** can access the data through the monitoring node **110**. In some examples, the monitoring node **110** is configured for browser-based accessibility. The browser-based interface can support well-known browsers. Alternatively, or in conjunction the entities **80** can obtain the information using one or more APIs through their device. The browser-based interface can include a website through which the data can be accessible. Although the website can be hosted by the monitoring node **110**, it can also be hosted at another location accessible through the network.

(47) The monitoring methodology can be used on a variety of different aircraft **20**. Aircraft **20** include but are not limited to manned aircraft, unmanned aircraft, manned spacecraft, unmanned spacecraft, manned rotorcraft, and unmanned rotorcraft, and combinations thereof.

(48) By the term “substantially” with reference to amounts or measurement values, it is meant that the recited characteristic, parameter, or value need not be achieved exactly. Rather, deviations or variations, including, for example, tolerances, measurement error, measurement accuracy limitations, and other factors known to those skilled in the art, may occur in amounts that do not preclude the effect that the characteristic was intended to provide.

(49) Further, the disclosure comprises examples according to the following clauses:

(50) Clause 1. A computing device to determine engine thrust of an aircraft **20** during taxiing, the computing device comprising: memory circuitry **112** configured to store computer-readable program code **115**; and processing circuitry **111** configured to execute the computer-readable program code **115** to cause the computing device to: identify an aircraft **20** on an airport surface **101**; calculate an acceleration of the aircraft **20** during the taxiing of the aircraft **20**; compare the acceleration with predetermined data from similar aircraft **20**; and based on the comparison, determine an engine thrust for the aircraft **20** while the aircraft **20** is taxiing.

(51) Clause 2. The computing device of Clause 1, wherein comparing the acceleration with the predetermined data from similar aircraft **20** comprises comparing the acceleration and the time period with acceleration data and time period data for other aircraft **20** each with a common manufacturer and model.

(52) Clause 3. The computing device of either of Clauses 1 or 2, wherein the processing circuitry **111** is further configured to determine a number of engines **21** on the aircraft **20** based on identified model of the aircraft **20**.

(53) Clause 4. The computing device of Clause 3, wherein for each of the engines **21**, the processing circuitry **111** is configured to determine whether each of the engines **21** is operating at a higher thrust level or a lower thrust level.

(54) Clause 5. The computing device of Clause 4, wherein the lower thrust level is a zero thrust level.

(55) Clause 6. The computing device of any of Clauses 1-5, wherein the processing circuitry **111** is configured to: determine a starting point of the aircraft **20** on the airport surface **101** when the aircraft **20** is stationary; determine an ending point that is positioned away from the starting point with the aircraft **20** moving along the airport surface **101** at the ending point; and calculate the acceleration based on a change in velocity between the starting and ending points and the time period for the aircraft **20** to move between the starting and ending points.

(56) Clause 7. The computing device of any of Clauses 1-6, wherein the processing circuitry **111** is configured to determine that the acceleration falls within a cluster **40** of data points **41** in the predetermined data and determine the engine thrust based on the cluster **40**.

(57) Clause 8. The computing device of any of Clauses 1-7, wherein the processing circuitry **111** is configured to determine a number of engines **21** and a thrust level of each of the engines **21**.

(58) Clause 9. The computing device of any of Clauses 1-8, wherein the predetermined data are stored at the monitoring node **110**.

(59) Clause 10. A computing device to determine engine thrust of an aircraft **20** during taxiing, the computing device comprising: memory circuitry **112** configured to store computer-readable program code **115**; and processing circuitry **111** configured to execute the computer-readable program code **115** to cause the computing device to: identify the aircraft **20** by model; identify a direction of travel of the aircraft **20**; determine an acceleration of the aircraft **20** over a time period during taxiing of the aircraft **20**; compare the acceleration and the time period with predetermined data of other aircraft **20** of the same model and the same direction of travel; determine that the acceleration and time period is within a cluster **40** of data points **41** in the predetermined data with the cluster **40** corresponding to an engine thrust for the aircraft **20**; and determine that the aircraft **20** is taxiing with a same number of engines **21** that are operating at same thrust levels as other aircraft from the corresponding cluster **40**.

(60) Clause 11. The computing device of Clause 10, wherein the processing circuitry **111** determines an engine thrust of the aircraft **20**.

(61) Clause 12. The computing device of either of Clauses 10 or 11, wherein the processing circuitry **111** calculates the acceleration between a first point where the aircraft **20** is stationary and a second point where the aircraft **20** is moving.

(62) Clause 13. The computing device of Clause 12, wherein the processing circuitry **111** calculates

the acceleration during the time period with a velocity of the aircraft **20** constantly increasing.

(63) Clause 14. The computing device of any of Claims 10-13, wherein the computing device is further configured to calculate the acceleration and the time period of the acceleration of the other aircraft and store the thrust levels as the data points in the predetermined data.

(64) Clause 15. A method of calculating engine thrust of an aircraft **20** that is taxiing along an airport surface **101** of an airport **100**, the method comprising: identifying an aircraft **20** that is on an airport surface **101**; calculating an acceleration over a time period of the aircraft **20** taxiing along the airport surface **101**; comparing the acceleration and the time period to data points **41** from other ones of the same aircraft **20** that each had an acceleration over a time period during taxiing; determining that the acceleration and the time period corresponds to a cluster **40** of the data points **41** with the cluster **40** corresponding to a number of engines **21** operating at one or more thrust levels; and determining that the aircraft is taxiing with the same number of engines **21** and the same thrust levels as the cluster **40**.

(65) Clause 16. The method of Claim 15, wherein identifying the aircraft **20** comprises identifying a model of the aircraft **20** from records maintained at the airport.

(66) Clause 17. The method of either of Claims 15 or 16, wherein calculating the acceleration of the aircraft **20** comprises determining a change in velocity of the aircraft **20** between first and second points along the airport surface **101** relative to the time period for the aircraft **20** to move between the first and second points with the velocity of the aircraft **20** at the first point being zero.

(67) Clause 18. The method of any of Claims 15-17, further comprising obtaining the data points **41** from one or more remote entities **80** that monitor other aircraft.

(68) Clause 19. The method of any of Claims 15-18, further comprising that each of the one or more engines **21** on the aircraft **20** has a full thrust level or a zero thrust level.

(69) Clause 20. The method of any of Claims 15-19, further comprising obtaining the data points **41** from previous aircraft **20** while taxiing at the airport.

(70) Clause A1. A method of determining engine thrust of an aircraft **20** during taxiing, the method comprising: identifying an aircraft **20** on an airport surface **101**; calculating an acceleration of the aircraft **20** during the taxiing of the aircraft **20**; comparing the acceleration with predetermined data from similar aircraft **20**; and based on the comparison, determining an engine thrust for the aircraft **20** while the aircraft **20** is taxiing.

(71) Clause A2. The method of Claim A1, wherein comparing the acceleration with the predetermined data from similar aircraft **20** comprises comparing the acceleration and the time period with acceleration data and time period data for other aircraft **20** each with a common manufacturer and model.

(72) Clause A3. The method of either of Claims A1 or A2, comprising determining a number of engines **21** on the aircraft **20** based on identified model of the aircraft **20**.

(73) Clause A4. The method of Claim A3, comprising for each of the engines **21**, determining whether each of the engines **21** is operating at a higher thrust level or a lower thrust level.

(74) Clause A5. The method of Claim A4, wherein the lower thrust level is a zero thrust level.

(75) Clause A6. The method of any of Claims A1-A5, comprising: determining a starting point of the aircraft **20** on the airport surface **101** when the aircraft **20** is stationary; determining an ending point that is positioned away from the starting point with the aircraft **20** moving along the airport surface **101** at the ending point; and calculating the acceleration based on a change in velocity between the starting and ending points and the time period for the aircraft **20** to move between the starting and ending points.

(76) Clause A7. The method of any of Claims A1-A6, comprising determining that the acceleration falls within a cluster **40** of data points **41** in the predetermined data and determine the engine thrust based on the cluster **40**.

(77) Clause A8. The method of any of Claims A1-A7, comprising determining a number of engines **21** and a thrust level of each of the engines **21**.

(78) Clause A9. The method of any of Claims A1-A8, wherein the predetermined data are stored at the monitoring node **110**.

(79) Clause A10. A method of determining engine thrust of an aircraft **20** during taxiing, the method comprising: identifying the aircraft **20** by model; identifying a direction of travel of the aircraft **20**; determining an acceleration of the aircraft **20** over a time period during taxiing of the aircraft **20**; comparing the acceleration and the time period with predetermined data of other aircraft **20** of the same model and the same direction of travel; determining that the acceleration and time period is within a cluster **40** of data points **41** in the predetermined data with the cluster **40** corresponding to an engine thrust for the aircraft **20**; and determining that the aircraft **20** is taxiing with a same number of engines **21** that are operating at same thrust levels as other aircraft from the corresponding cluster **40**.

(80) Clause A11. The method of Claim A10, comprising determining an engine thrust of the aircraft **20**.

(81) Clause A12. The method of either of Claims A10 or A11, comprising calculating the acceleration between a first point where the aircraft **20** is stationary and a second point where the aircraft **20** is moving.

(82) Clause A13. The method of Claim A12, comprising calculating the acceleration during the time period with a velocity of the aircraft **20** constantly increasing.

(83) Clause A14. The method of any of Claims A10-13, comprising calculating the acceleration and the time period of the acceleration of the other aircraft and store the thrust levels as the data points in the predetermined data.

(84) Clause A15. A method of calculating engine thrust of an aircraft **20** that is taxiing along an airport surface **101** of an airport **100**, the method comprising: identifying an aircraft **20** that is on an airport surface **101**; calculating an acceleration over a time period of the aircraft **20** taxiing along the airport surface **101**; comparing the acceleration and the time period to data points **41** from other ones of the same aircraft **20** that each had an acceleration over a time period during taxiing; determining that the acceleration and the time period corresponds to a cluster **40** of the data points **41** with the cluster **40** corresponding to a number of engines **21** operating at one or more thrust levels; and determining that the aircraft is taxiing with the same number of engines **21** and the same thrust levels as the cluster **40**.

(85) Clause A16. The method of Claim A15, wherein identifying the aircraft **20** comprises identifying a model of the aircraft **20** from records maintained at the airport.

(86) Clause A17. The method of either of Claims A15 or A16, wherein calculating the acceleration of the aircraft **20** comprises determining a change in velocity of the aircraft **20** between first and second points along the airport surface **101** relative to the time period for the aircraft **20** to move between the first and second points with the velocity of the aircraft **20** at the first point being zero.

(87) Clause A18. The method of any of Claims A15-A17, further comprising obtaining the data points **41** from one or more remote entities **80** that monitor other aircraft.

(88) Clause A19. The method of any of Claims A15-A18, further comprising that each of the one or more engines **21** on the aircraft **20** has a full thrust level or a zero thrust level.

(89) Clause A20. The method of any of Claims A15-A19, further comprising obtaining the data points **41** from previous aircraft **20** while taxiing at the airport.

(90) Clause A21. A computer program comprising computer program instructions that, when executed by computer processing circuitry **111**, cause the computer processing circuitry **111** to perform the method of any of Claims A1-A20.

(91) Clause A22. A computer-readable storage medium having stored therein the computer program of Claim A21.

(92) Clause A23. A computing device for determining engine thrust of an aircraft **20** during taxiing, the computing device comprising: memory circuitry **112** configured to store the computer program of Claim A21; and computer processing circuitry **111** configured to execute the computer program

thereby causing the computer processing circuitry 111 to perform the method of any of Claims A1-A20.

Claims

1. A method of determining engine thrust of an aircraft during taxiing, the aircraft comprising a number of engines, the method comprising: identifying an aircraft on an airport surface; calculating an acceleration of the aircraft during the taxiing of the aircraft; comparing the acceleration of the aircraft to acceleration data of other aircraft with an equal number of engines, the acceleration data comprising clusters of data points with a number of clusters being the same as the number of engines and with each of the clusters of data points corresponding to a different engine thrust; determining that the acceleration of the aircraft is aligned within one of the cluster of data points; determining an engine thrust for the aircraft based on the cluster in which the acceleration coincides.
2. The method of claim 1, wherein the comparing the acceleration with the acceleration data from similar aircraft comprises comparing the acceleration and a time period during taxiing of the aircraft with acceleration data and time period data for other aircraft each with a common manufacturer and model.
3. The method of claim 1, comprising determining the number of engines on the aircraft based on an identified model of the aircraft.
4. The method of claim 3, comprising identifying a model of the aircraft from records maintained at the airport.
5. The method of claim 3, wherein for each of the engines, determining whether each of the engines is operating at a higher thrust level or a lower thrust level.
6. The method of claim 5, wherein the lower thrust level is a zero thrust level.
7. The method of claim 1, further comprising: determining a starting point of the aircraft on the airport surface when the aircraft is stationary; determining an ending point that is positioned away from the starting point with the aircraft moving along the airport surface at the ending point; and calculating the acceleration based on a change in velocity between the starting and ending points and a time period for the aircraft to move between the starting and ending points.
8. The method of claim 1, comprising determining the number of engines and a thrust level of each of the engines.
9. The method of claim 1, wherein the acceleration data are stored at a monitoring node.
10. The method of claim 9, wherein the monitoring node is selected from the group consistently essential of a control tower of an airport that monitors the overall operation of the airport, an airline that schedules the arrival and departure of aircraft, a flight tracker that monitors the status of flights, and a governmental unit.
11. A computer program product comprising a non-transitory computer-readable medium having computer-executable program code instructions that, when executed by a computing device, cause the computing device to: identify an aircraft on an airport surface and determine a number of engines on the aircraft; calculate an acceleration of the aircraft during the taxiing of the aircraft; compare the acceleration with predetermined data from similar aircraft with an equal number of engines, the data comprising clusters of data points with each of the clusters corresponding to a different engine thrust; determine that the acceleration of the aircraft corresponds to one of the clusters of data points; and based on the comparison, determine the engine thrust for the aircraft while the aircraft is taxiing.
12. The computer program product of claim 11, wherein the computing device is further configured to determine that the aircraft is taxiing with the same number of engines and the same thrust levels as the cluster.
13. The computer program product of claim 11, wherein the computing device is further configured

to identify a model of the aircraft from records maintained at an airport.

14. The computer program product of claim 11, wherein the calculating the acceleration of the aircraft comprises determining a change in velocity of the aircraft between first and second points along the airport surface relative to a time period for the aircraft to move between the first and second points with a velocity of the aircraft at the first point being zero.

15. The computer program product of claim 11, wherein the computing device is further configured to obtain the data points from one or more remote entities that monitor other aircraft.

16. A computing device for determining engine thrust of an aircraft during taxiing, the computing device comprising: memory circuitry configured to store computer-readable program code; and processing circuitry configured to execute the computer-readable program code to cause the computing device to: identify the aircraft by model and a number of engines of the aircraft; identify a direction of travel of the aircraft; determine an acceleration of the aircraft over a time period during taxiing of the aircraft; compare the acceleration and the time period with predetermined data of other aircraft of the same model, the same direction of travel, and with the same number of engines and with the other aircraft having an acceleration over a time period during taxiing; determine that the acceleration and time period is within a cluster of data points in the predetermined data with the cluster corresponding to an engine thrust for the aircraft; and determine that the aircraft is taxiing with a same number of engines that are operating at same thrust levels as other aircraft from the corresponding cluster.

17. The computing device of claim 16, wherein the processing circuitry calculates the acceleration between a first point where the aircraft is stationary and a second point where the aircraft is moving.

18. The computing device of claim 17, wherein the processing circuitry calculates the acceleration during the time period with a velocity of the aircraft constantly increasing.

19. The computer program product of claim 11, wherein the computing device is further configured to determine whether the engines on the aircraft are operating at a high thrust level or a low thrust level.

20. The computer program product of claim 19, wherein the computing device is further configured to determine whether the engines are at a zero-thrust level.
